Fitted Q Evaluation Without Bellman Completeness via Stationary Weighting

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Abstract

Fitted Q-evaluation (FQE) is a foundational method for off-policy evaluation in reinforcement learning, but existing theory typically relies on Bellman completeness of the function class, a condition often violated in practice. We trace this reliance to a fundamental *norm mismatch*: the Bellman operator is γ-contractive in the L^2 norm induced by the target policy's stationary distribution, whereas standard FQE fits Bellman regressions under the behavior distribution. To resolve this mismatch, we reweight each Bellman regression step by an estimate of the stationary density ratio, inspired by emphatic weighting in temporal-difference learning. This makes the update behave as if it were performed under the target stationary distribution, restoring contraction without Bellman completeness while preserving the simplicity of regression-based evaluation. Illustrative experiments, including Baird's classical counterexample, show that stationary weighting can stabilize FQE under off-policy sampling.

1. Introduction

Fitted Q-evaluation (FQE), a form of approximate Bellman iteration for policy evaluation, is a standard tool in reinforcement learning (Tsitsiklis & Van Roy, 1996; Gordon, 1995; Lagoudakis & Parr, 2003; Munos, 2005; Munos & Szepesvári, 2008; Farahmand et al., 2010; Bertsekas, 2011). Its modularity makes it particularly appealing in offline and batch RL: each iteration constructs one-step Bellman targets for policy evaluation and regresses them onto a function class, allowing practitioners to plug in off-the-shelf supervised learners (Fujimoto et al., 2019; Agarwal et al., 2021). At the population level, this corresponds to Bellman iteration for the policy-evaluation operator, which is contractive and has the policy-value Q-function as its fixed point.

Despite its practical appeal, FQE can be fragile under function approximation and distribution shift, as reflected in analyses that rely on Bellman completeness (Gordon, 1995; Baird et al., 1995; Tsitsiklis & Van Roy, 1996; Munos, 2003; Munos & Szepesvári, 2008; Melo et al., 2008; Bertsekas, 2011; Scherrer, 2014; Fan et al., 2020; Hu et al., 2025). Bellman completeness requires that the function class be closed under the Bellman operator, so that each one-step Bellman target remains representable within the class. When this fails, projection errors can accumulate across iterations and may even grow exponentially. This instability is also reflected in recent information-theoretic hardness results for misspecified or offline fitted value and Q-evaluation (Chen & Jiang, 2019; Foster et al., 2021; Wang et al., 2021; Amortila et al., 2020; Mou et al., 2020). At the same time, projected fixed points can exist, and in some cases iterative methods can converge to them, even without Bellman completeness (Meyn, 2024; Lim & Lee, 2025; Chen et al., 2020). However, such results generally do not recover the classical linear convergence guarantees for FQE. These limitations have motivated adversarial Q-learning methods that relax completeness through saddle-point formulations (Dai et al., 2018; Uehara et al., 2020; Xie & Jiang, 2020; Xie et al., 2021; Uehara et al., 2021; Zhan et al., 2022; Uehara et al., 2023; Jin et al., 2021; Xie & Jiang, 2021), but at the cost of minimax objectives, auxiliary critic classes, and a departure from standard supervised learning with its many off-the-shelf algorithms.

A norm-mismatch perspective. *The core instability of standard FQE arises from a mismatch between the regression norm and the norm in which the Bellman operator is contractive* . For policy evaluation, the Bellman operator $\mathcal{T}$ is γ-contractive in the L^2 norm induced by the target policy's stationary distribution, whereas standard FQE performs regression in the behavior (sampling) norm, where such contraction need not hold (Bertsekas & Tsitsiklis, 1996; Hallak et al., 2016; Patterson et al., 2022). Classical guarantees control this mismatch only through realizability and (approximate) Bellman completeness.

Related instability issues have long been recognized in the TD-learning literature (Tsitsiklis & Van Roy, 1996; Precup et al., 2001; Yu & Bertsekas, 2012). TD-style updates are naturally stable under on-policy stationary sampling but can become unstable off policy. This motivated emphatic weighting (Mahmood et al., 2015), which reweights TD updates to restore a contractive projected Bellman operator under off-policy sampling (Yu, 2015; Sutton et al., 2016; Hallak et al., 2016; Hallak & Mannor, 2017). Most of this literature studies TD-style methods in finite-state settings with linear function approximation, with more recent exten-

sions to nonlinear settings (Jiang et al., 2021; Patterson et al., 2022). Whether similar ideas can stabilize regression-based fitted methods such as FQE, particularly under nonlinear function approximation, remains much less understood.

We show that FQE admits an analogous weighting-based correction tailored to the fitted setting. Because FQE proceeds by regressing Bellman targets under the behavior distribution, the appropriate remedy is to modify only the regression geometry: reweight each step by stationary-density-ratio weights. This restores contraction under misspecification while preserving the simplicity of fitted Bellman iteration and avoiding saddle-point reformulations.

1.1. Our main contributions.

- We formalize *stationary-weighted* FQE for offline policy evaluation, reweighting each Bellman regression step by stationary-density-ratio weights to correct the regression geometry induced by the behavior distribution.

- We develop a novel finite-sample analysis for stationary-weighted FQE that separates finite-iteration, statistical, approximation, and weight-estimation errors, extending the inexact fixed-point framework of Munos & Szepesvári (2008) to approximate weighting. We further show that under approximate Bellman completeness, the effect of weight-estimation error is attenuated through the inherent Bellman error.

Scope and limitations. Our focus in this paper is primarily conceptual and theoretical, complemented by proof-of-concept experiments. We bring the geometric intuition behind emphatic weighting in off-policy TD learning to regression-based fitted value methods, showing how stationary reweighting can stabilize fitted Q-evaluation. This contribution is most relevant under nonlinear function approximation, since linear settings are comparatively well understood (Melo & Ribeiro, 2007; Yang & Wang, 2020; Jin et al., 2023).

The main limitations are the need for sufficient stationary-behavior coverage and the difficulty of density-ratio estimation. While there is a large literature on stationary density-ratio estimation (see Section 3.2) and emphatic weights more broadly (Jiang et al., 2021; Patterson et al., 2022), obtaining stable and accurate weights under nonlinear function approximation remains challenging in practice (Zhang & Whiteson, 2022). Our goal is not to solve that practical problem, but to isolate stationary reweighting as a stabilizing mechanism for regression-based fitted value methods under distribution mismatch.

1.2. Related Work

Relation to classical analyses. Classical analyses of fitted value iteration and fitted Q-evaluation typically follow one of two paths. One assumes that the Bellman operator is already contractive in the regression norm, possibly with a degraded contraction factor (Gordon, 1995; Tsitsiklis & Van Roy, 1996; Bertsekas, 2011). The other, exemplified by Munos & Szepesvári (2008), controls how approximation and estimation errors propagate across iterations. Such guarantees often rely on strong assumptions on how the Bellman operator interacts with the function class, including Bellman-completeness-type conditions (Munos & Szepesvári, 2008; Chen & Jiang, 2019; Zanette et al., 2020), linear structure (Melo & Ribeiro, 2007; Duan et al., 2020; Yang & Wang, 2020; Jin et al., 2023), or modified algorithms based on learned representations (Chang et al., 2022), state-space discretization (Peng & Williams, 1993; Van Roy, 2006; Xie & Jiang, 2021), or minimax Bellman-error objectives (Uehara et al., 2020; Xie & Jiang, 2020). In contrast, motivated by emphatic weighting in TD learning (Mahmood et al., 2015), we study a reweighting that makes the projected Bellman operator contractive, thereby yielding linear convergence.

Weighted temporal-difference methods. Emphatic weighting and related off-policy TD methods use reweighting to stabilize projected Bellman updates under distribution mismatch (Mahmood et al., 2015; Sutton et al., 2016; Hallak et al., 2016; Patterson et al., 2022), building on earlier work on weighted TD learning (Precup et al., 2001) and weighted Bellman equations (Yu & Bertsekas, 2012; Yu et al., 2018). Relatedly, Patterson et al. (2022) establishes stability of projected Bellman updates under the stationary norm for emphatic TD learning. These works are closely related in spirit to ours: in both settings, weighting is used to improve the contraction geometry under off-policy sampling.

What remains less understood is how these ideas extend to regression-based fitted methods such as FQE, especially under nonlinear function approximation and general state spaces. In this work, we isolate the role of weighting relative to the Bellman-completeness requirement in FQE. Unlike prior emphatic-weighting analyses, which focus on TD-style recursions and typically assume contraction under the weights used by the method, we study fitted Bellman regression, require contraction only under the true stationary weights, and explicitly quantify the effect of weight-estimation error.

Our setting therefore requires a different analysis from prior TD-learning work and also departs from most of that literature by focusing on nonlinear function approximation. The resulting picture is also different: under Bellman completeness, the choice of weights is irrelevant, whereas outside this regime appropriate weighting can restore contraction. More generally, our analysis shows that when the inherent

Bellman error is small, the effect of weight-estimation error is correspondingly attenuated, so approximate Bellman completeness also stabilizes the fitted updates.

Minimax and pessimistic Bellman-error methods. A parallel line of work controls worst-case Bellman error using adversarial or pessimistic critics (Uehara et al., 2020; 2021; Jin et al., 2021; Xie & Jiang, 2021; Di et al., 2023; Amortila et al., 2024). These methods optimize saddle-point objectives, introduce auxiliary critic classes, and rely on assumptions such as partial coverage and dual realizability. They are related to our approach in that the critic class often implicitly approximates discounted occupancy-ratio weights (see, e.g., Section 6 of Xie & Jiang, 2020). Unlike these methods, however, we retain the standard regression form of FQE and modify only the weighting of the regression step.

Stationary density-ratio estimation. A large literature studies density-ratio estimation for off-policy evaluation, including stationary-distribution estimators (Glynn & Henderson, 1998; Paulin, 2015; Meggendorfer, 2023) and discounted-occupancy or stationary-discounted ratios (Nachum et al., 2019a; Zhang et al., 2020; Lee et al., 2022; Uehara et al., 2020; Nachum et al., 2019b; Mao et al., 2024; van der Laan et al., 2025a). DICE-style methods use such ratios to estimate average policy values. Our work uses them to stabilize FQE.

2. Problem Setup

2.1. Preliminaries

We consider a discounted Markov decision process (MDP) with continuous state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition kernel $P(\cdot \mid s,a)$, reward function $r_0(s,a)$, and discount factor $\gamma \in [0,1)$. A target policy $\pi(a \mid s)$ induces the Markov operator

$$(\pi P Q)(s,a) := \mathbb{E}_\pi[Q(S',A') \mid S=s,\ A=a]\,,$$

where $S' \sim P(\cdot \mid s,a)$ and $A' \sim \pi(\cdot \mid S')$. The associated policy-evaluation Bellman operator is (Bellman, 1952; 1966)

$$\mathcal{T}Q := r_0 + \gamma\,(\pi P Q).$$

The Q-function, or state-action value function, is

$$Q^\star(s,a) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty}\gamma^t r_0(S_t,A_t)\;\middle|\; S_0=s,\ A_0=a\right],$$

and is the unique fixed point of the Bellman operator, satisfying $Q^\star = \mathcal{T}Q^\star$.

We observe i.i.d. transitions $\mathcal{D}_n = \{(S_i,A_i,R_i,S_i')\}_{i=1}^n$, where $(S,A)\sim\nu_b$, $S'\sim P(\cdot \mid S,A)$, and $R = r_0(S,A)+\varepsilon$ with $E[\varepsilon \mid S,A]=0$. Here ν_b denotes the behavior distribution on $\mathcal{S}\times\mathcal{A}$. Following Xie & Jiang (2021), we focus on i.i.d. transition sampling for simplicity; the analysis extends to dependent trajectory data under appropriate mixing conditions.

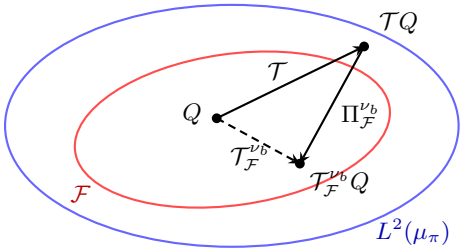

Figure 1. Norm mismatch: $Q \in \mathcal{F}$ (red), $\mathcal{T}Q$ leaves $\mathcal{F}$ but lies in $L^2(\mu_\pi)$ (blue), and $\Pi_{\mathcal{F}}^{\nu_b}$ projects it back under the behavior-norm $L^2(\nu_b)$. The projection $\mathcal{T}_{\mathcal{F}}^{\nu_b} = \Pi_{\mathcal{F}}^{\nu_b}\mathcal{T}$ need not be contractive.

For any policy π and function f, we write

$$(\pi f)(s) := \sum_{a\in\mathcal{A}} \pi(a \mid s)\, f(s,a).$$

For any distribution ν on $\mathcal{S}\times\mathcal{A}$, let $L^2(\nu)$ denote the space of measurable functions $f:\mathcal{S}\times\mathcal{A}\to\mathbb{R}$ such that $\int f(s,a)^2\,\nu(ds,da) < \infty$, equipped with norm

$$\|f\|_{2,\nu} := \left(\int_{\mathcal{S}\times\mathcal{A}} f(s,a)^2\,\nu(ds,da)\right)^{1/2}.$$

Stationary distributions. A distribution μ_π on $\mathcal{S}\times\mathcal{A}$ is *stationary* for (π,P) if it is preserved by one-step evolution under policy π and transition kernel P, that is, $\mu_\pi = \mu_\pi \pi P$. Intuitively, for an ergodic Markov chain induced by (π,P), μ_π is the long-run state-action distribution (Puterman, 1990).

2.2. Standard FQE and Norm Mismatch

Classical dynamic programming computes the Q-function $Q^\star$ by iterating the Bellman operator $\mathcal{T}$, viewing value estimation as a Picard iteration toward the fixed point $Q^\star = \mathcal{T}Q^\star$ (Bellman, 1966). In tabular settings this can be done exactly, but in larger or continuous state spaces one must combine Bellman updates with function approximation over a class $\mathcal{F}$ (Sutton et al., 1998). Standard fitted Q-evaluation (FQE) does so by constructing iterates

$$\widehat{Q}^{(k+1)} \in \arg\min_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^n \big(R_i + \gamma\,(\pi\widehat{Q}^{(k)})(S_i') - f(S_i,A_i)\big)^2,$$

that is, by repeatedly fitting least-squares regressions to one-step Bellman targets under the behavior distribution ν_b. In the population limit, this corresponds to iterating the projected Bellman operator $\mathcal{T}_{\mathcal{F}}^{\nu_b} := \Pi_{\mathcal{F}}^{\nu_b}\mathcal{T}$, where $\Pi_{\mathcal{F}}^{\nu_b}$ denotes the $L^2(\nu_b)$ projection onto $\mathcal{F}$.

Classical analyses (Munos & Szepesvári, 2008) effectively assume that iterating $\mathcal{T}_{\mathcal{F}}^{\nu_b}$ closely tracks iteration of $\mathcal{T}$ in

an appropriate norm, so that iterating $\mathcal{T}_{\mathcal{F}}^{\nu_b}$ converges to the unique fixed point $Q^\star$ of $\mathcal{T}$ up to an inherent Bellman error. A standard sufficient condition for convergence is *Bellman completeness* of $\mathcal{F}$, meaning $\mathcal{T}f \in \mathcal{F}$ for all $f \in \mathcal{F}$; in this case, $\Pi_{\mathcal{F}}^{\nu_b}\mathcal{T}f = \mathcal{T}f$ on $\mathcal{F}$, so the projection does not interfere with the contraction of $\mathcal{T}$ (Chen & Jiang, 2019).

When $\mathcal{F}$ is not Bellman complete, the projected Bellman operator $\mathcal{T}_{\mathcal{F}}^{\nu_b}$ need not be contractive, and FQE may fail to converge or behave unstably. The issue is geometric: $\Pi_{\mathcal{F}}^{\nu_b}$ is nonexpansive in $L^2(\nu_b)$, whereas $\mathcal{T}$ is often analyzed as contracting in a different norm, such as $\|\cdot\|_\infty$. These norms need not be comparable (Figure 1). Even when the iterates do converge, convergence may be slow, and the limiting behavior-projected solution can be far from the true fixed point $Q^\star$ (Tsitsiklis & Van Roy, 1996; De Farias & Van Roy, 2000; Patterson et al., 2022).

2.3. On-Policy Stability and Resolving Norm Mismatch

A natural question is which norm gives the right geometry for Bellman iteration. A classical insight from RL is that under on-policy sampling from the target policy's stationary distribution μ_π, Bellman updates can be stable even under function approximation (Tsitsiklis & Van Roy, 1996; Hallak et al., 2016; Patterson et al., 2022). The reason is that projection then occurs in $L^2(\mu_\pi)$, the same geometry in which the Bellman operator contracts, rather than in the behavior norm $L^2(\nu_b)$. As a result, the projected Bellman operator inherits contraction. This suggests a simple fix for off-policy FQE: reweight each regression step so that it behaves as if the data were sampled from μ_π, paralleling the role of emphatic weighting in TD learning.

To formalize this, let $\mathcal{F} \subseteq L^2(\mu_\pi)$ be convex, and let $\overline{\mathcal{F}}$ denote its closure in $L^2(\mu_\pi)$. We define the projected Bellman operator in $L^2(\mu_\pi)$ by

$$\mathcal{T}_{\mathcal{F}} := \Pi_{\mathcal{F}}\mathcal{T}, \qquad \Pi_{\mathcal{F}} g := \arg\min_{f\in\overline{\mathcal{F}}} \|f-g\|_{2,\mu_\pi}. \tag{1}$$

C1 Existence of stationary distribution. The MDP induced by (π, P) admits a stationary distribution μ_π.

C2 Convex function class. $\mathcal{F}$ is a convex subset of $L^2(\mu_\pi)$.

Lemma 1 (Contraction of Bellman operator)**.** *Assume C1. Then the Bellman operator $\mathcal{T}$ is a γ-contraction on $L^2(\mu_\pi)$:*

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_{2,\mu_\pi} \le \gamma\|Q-Q'\|_{2,\mu_\pi}, \qquad \forall\, Q, Q' \in L^2(\mu_\pi).$$

If C2 also holds, then the projected operator $\mathcal{T}_{\mathcal{F}}$ is a γ-contraction on $\mathcal{F}$.

Condition C1 holds under mild ergodicity conditions (Puterman, 1990). Condition C2 is standard and ensures that $\Pi_{\mathcal{F}}$ is well defined and nonexpansive (Munos & Szepesvári, 2008; Patterson et al., 2022). It is satisfied by many common classes, including linear and RKHS-based function classes.

Algorithm 1 Stationary-Weighted Fitted Q-Evaluation

input Offline data $\mathcal{D}_n$, target policy π, function class $\mathcal{F}$, iterations K, discount γ

```
# Estimate stationary density ratio
```

$\widehat{d}_\pi \leftarrow$ estimator of d_π

Initialize $\widehat{Q}^{(0)} \in \mathcal{F}$

for $k = 0, \dots, K-1$ **do**

```
# Construct targets and weights
```

$Y_i^{(k)} \leftarrow R_i + \gamma\,(\pi\widehat{Q}^{(k)})(S_i')$

$w_i \leftarrow \widehat{d}_\pi(S_i, A_i)$

```
# Stationary-weighted regression
```

$\widehat{Q}^{(k+1)} \leftarrow \arg\min_{f\in\mathcal{F}} \sum_{i=1}^n w_i\big\{Y_i^{(k)} - f(S_i,A_i)\big\}^2$

end for

output $\widehat{Q}^{(K)}$

3. Stationary-Weighted Fitted Q-Evaluation

3.1. Algorithm

Motivated by the on-policy stability of FQE, we replace the behavior-weighted regression loss in standard FQE with a weighted loss targeting the stationary $L^2(\mu_\pi)$ geometry. As we show in Section 4.2, our theory only requires an *admissible* weighting function, of which the stationary density ratio is one natural choice.

Define the stationary density ratio for (π, P) by

$$d_\pi(s,a) := \frac{d\mu_\pi}{d\nu_b}(s,a),$$

assuming $\mu_\pi \ll \nu_b$. Given an estimate $\widehat{d}_\pi$, let $w_i := \widehat{d}_\pi(S_i, A_i)$. We then define the *stationary-weighted* FQE update by

$$\widehat{Q}^{(k+1)} = \arg\min_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^n w_i\Big\{R_i + \gamma(\pi\widehat{Q}^{(k)})(S_i') - f(S_i,A_i)\Big\}^2.$$

When $\widehat{d}_\pi = d_\pi$, this update corresponds to projection in $L^2(\mu_\pi)$.

Algorithm 1 summarizes the procedure, and Section 4 provides finite-sample guarantees. The approach is compatible with generic function approximators, including neural networks and boosted ensembles. Relative to standard FQE, the main additional computational cost is a one-time density-ratio estimation step, whose cost depends on the particular weight-estimation procedure used.

Interpretation of the stationary norm. Beyond being the norm in which contraction holds, $L^2(\mu_\pi)$ is a natural geometry for policy evaluation because it weights prediction error by the target policy's long-run visitation frequencies. Since μ_π depends only on (π, P), this norm is invariant to the behavior policy and the initial-state distribution, and

thus reflects long-run prediction performance rather than the particulars of the data-collection process (Puterman, 2014; Levin & Peres, 2017).

3.2. Estimation of Stationary Ratios

A key component of stationary-weighted FQE is estimating the stationary density ratio. This is challenging in its own right and typically leads to a minimax or saddle-point problem, as in DICE-style methods (Nachum et al., 2019a; Zhang et al., 2020; Lee et al., 2021) and related minimax weight-estimation approaches (Uehara et al., 2020; Wang et al., 2023). Stationary ratios are closely related to the discounted occupancy ratios widely used in RL, and arise as their limit as $\gamma \to 1$ under standard ergodicity conditions.

Since $d_\pi \nu_b = \mu_\pi$ is a stationary distribution for (π, P), the stationary ratio $d_\pi = d\mu_\pi / d\nu_b$ satisfies the moment condition

$$\mathbb{E}\big[d_\pi(S,A)\,\big(g(S,A) - g(S',A')\big)\big] = 0 \quad \text{for all } g, \tag{2}$$

where $(S,A) \sim \nu_b$, $S' \sim P(\cdot \mid S,A)$, and $A' \sim \pi(\cdot \mid S')$.

Minimax and DICE-style saddle-point methods enforce stationarity through a variational formulation and learn the ratio using paired weight and critic classes. A representative empirical objective is

$$\arg\min_{d \in \mathcal{D}} \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i)\,\big(g(S_i,A_i) - g(S_i', A_i')\big) - \frac{1}{2n} \sum_{i=1}^{n} g(S_i, A_i)^2 \right\} + \left(\frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i) - 1 \right)^2,$$

where $A_i' \sim \pi(\cdot \mid S_i')$, and the final term enforces normalization. When the critic class $\mathcal{G}$ is linear or an RKHS, the inner maximization can often be expressed in closed form (Olivas-Martinez & Rotnitzky, 2025). In practice, such structured critic classes can yield simpler and more stable estimators than highly flexible saddle-point methods, while still producing useful weights; see, for example, approximate balancing weights (Wang et al., 2023). Theoretical analyses of minimax weight estimators are given by Dikkala et al. (2020); Uehara et al. (2020); Wang et al. (2023).

Importantly, weighting can help even without accurate stationary-ratio estimation. Because the weights define only the regression norm, even misspecified weights may improve stability by better aligning the update with the on-policy geometry. See Patterson et al. (2022) and Section 4.4 for further discussion.

4. Theoretical Guarantees

We study the ERM version of stationary-weighted FQE in Algorithm 1. Our proof strategy follows the inexact Picard iteration framework of (Munos & Szepesvári, 2008), with a new analysis of the per-iteration error under estimated weights. The main technical challenge is that the Bellman operator contracts only under the true weights, whereas the procedure uses estimated weights that need not contract.

The analysis proceeds as follows. Section 4.1 isolates the function-approximation error by characterizing the FQE target as the fixed point of a projected Bellman equation. Section 4.2 introduces admissible weight functions, with the stationary density ratio as a concrete example, and shows how weighting relaxes Bellman completeness. Section 4.3 presents our main technical lemma, which controls the per-iteration regression error and quantifies how weight-estimation error interacts with the inherent Bellman error. Our main result is presented in Section 4.4.

4.1. Projected Bellman Fixed Point

The population target of stationary-weighted FQE is the fixed point $Q^\star_{\mathcal{F}}$ of the projected Bellman operator

$$\mathcal{T}_{\mathcal{F}} := \Pi_{\mathcal{F}} \mathcal{T}.$$

If $Q^\star \in \mathcal{F}$, then $Q^\star_{\mathcal{F}} = Q^\star$. When realizability fails, the next result shows that $Q^\star_{\mathcal{F}}$ lies within a factor $(1-\gamma)^{-1}$ of the best $L^2(\mu_\pi)$ approximation to $Q^\star$ in $\mathcal{F}$.

Theorem 1 (Function approximation error)**.** *Under C1–C2,*

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma} \inf_{f \in \mathcal{F}} \|f - Q^\star\|_{2,\mu_\pi}.$$

4.2. Admissible Weighting Functions

Our theory does not require the exact stationary density ratio. Instead, it only assumes the existence of a weighting function that induces the same projection geometry on $\mathcal{F}$ as the stationary norm. This requirement is strictly weaker than Bellman completeness, $\mathcal{T}(\mathcal{F}) \subseteq \mathcal{F}$, under which the projection geometry is irrelevant and any weighting function is admissible.

C3 Bellman residual orthogonality under weighting. There exists nonnegative $d_\pi \in L^2(\nu_b)$ such that, for all $f, Q \in \mathcal{F}$,

$$\mathbb{E}_{\nu_b}\big[d_\pi(S,A) f(S,A) \{\mathcal{T}Q - \Pi_{\mathcal{F}}\mathcal{T}Q\}(S,A)\big] = 0.$$

Condition C3 is the first-order optimality condition ensuring that the projected update

$$\mathcal{T}_{\mathcal{F}}(Q^{(k)}) = \arg\min_{f \in \mathcal{F}} \|\mathcal{T}(Q^{(k)}) - f\|_{2,\mu_\pi}$$

can equivalently be computed by d_π-weighted regression under the behavior distribution ν_b:

$$\arg\min_{f \in \mathcal{F}} \mathbb{E}_{\nu_b}\Big[d_\pi(S,A)\{\mathcal{T}(Q^{(k)})(S,A) - f(S,A)\}^2\Big].$$

Equivalently, Condition C3 requires the Bellman residual $\mathcal{T}Q - \Pi_{\mathcal{F}}\mathcal{T}Q$ to be orthogonal to $\mathcal{F}$ in $L^2(d_\pi \nu_b)$. Under Bellman completeness, this holds automatically because the residual is zero. When d_π is the stationary density ratio, Condition C3 requires the behavior distribution to provide sufficient coverage of the stationary distribution (Xie et al., 2022), thereby avoiding existing hardness results (Foster et al., 2021).

Hereafter, let $\hat{d}_\pi$ denote any estimator of a weighting function d_π satisfying Condition C3. A natural choice for d_π is the stationary density ratio. If $\mathcal{F}$ is Bellman complete, one may take $d_\pi = \hat{d}_\pi$, in which case our results recover the classical guarantees.

Relation to dual realizability in minimax Q-learning. Condition C3 is related to, but distinct from, the assumptions used in minimax, saddle-point, and other moment-based approaches to Q-function estimation (Uehara et al., 2020; Xie et al., 2021; Uehara et al., 2021; 2023; Zanette & Wainwright, 2022). Both frameworks require a form of coverage of the target distribution: minimax methods typically assume coverage of the target discounted occupancy measure, whereas we assume overlap with the target stationary distribution. These notions are closely related in long-horizon settings, since the discounted occupancy ratio converges to the stationary ratio d_π as $\gamma \to 1$ under standard ergodicity conditions (Meyn & Tweedie, 1993). The main difference is that minimax methods also typically require *dual realizability*, meaning that the relevant ratio or moment equation must lie in, or be well approximated by, a critic class (see Section 6 of Xie et al. (2021)). By contrast, our approach does not require critic-class realizability, only sufficiently accurate estimation of an admissible weight.

4.3. One-step regression error under approximate weighting

This section gives the key ingredient for the finite-sample analysis. We first state a standard deterministic inexact Picard iteration bound, which reduces the problem to controlling the per-iteration regression errors. We then present our main technical lemma, which bounds these errors under approximate weighting.

Lemma 2 (Approximate Picard iteration error bound)**.** *Assume C1–C2. If $\{\eta_k\}_{k\geq 1}$ satisfies*

$$\|\hat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}\hat{Q}^{(k-1)}\|_{2,\mu_\pi} \leq \eta_k \qquad \textit{for all } k,$$

then for any $K \geq 1$,

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \leq \gamma^K \, \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \sum_{j=1}^{K} \gamma^{K-j} \, \eta_j.$$

To apply Lemma 2, it remains to control the per-iteration regression errors $\|\widehat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}\widehat{Q}^{(k-1)}\|_{2,\mu_\pi}$. This is the main new step in the analysis and the point at which approximate weighting enters.

We measure regression complexity via the extended class

$$\mathcal{F}_{\text{ext}} := \mathcal{F} \cup \{\, \pi f : f \in \mathcal{F} \,\}.$$

For $\varepsilon > 0$, let $N(\varepsilon, \mathcal{F}_{\text{ext}}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}_{\text{ext}}$ under $L^2(Q)$, and define the uniform entropy integral (Van Der Vaart & Wellner, 1996)

$$\mathcal{J}(\delta, \mathcal{F}_{\text{ext}}) := \int_0^\delta \sup_Q \sqrt{\log N(\varepsilon, \mathcal{F}_{\text{ext}}, L^2(Q))}\, d\varepsilon,$$

where the supremum is over discrete distributions Q. The statistical accuracy of each regression step is then governed by the critical radius

$$\delta_n := \inf\Big\{\delta > 0 : \frac{\mathcal{J}(\delta, \mathcal{F}_{\text{ext}})}{\sqrt{n}\,\delta^2} \leq 1\Big\}.$$

We impose the following mild technical conditions, several of which are standard and can be relaxed with more refined analysis.

C4 Uniform boundedness. The quantities $\hat{d}_\pi$, d_π, R, and $\{\hat{Q}^{(k)}, \mathcal{T}_{\mathcal{F}}\hat{Q}^{(k)}\}_{k=0}^K$ are uniformly bounded by a constant $M < \infty$.

C5 Stationary–weighted norm equivalence. There exist $1 \leq c < \infty$ such that for all $f \in L^2(\mu_\pi)$,

$$c^{-1}\|f\|_{2,\mu_\pi} \;\leq\; \|f\|_{2,\, d_\pi \nu_b} \;\leq\; c\|f\|_{2,\mu_\pi}.$$

C6 Sample splitting. The estimator $\hat{d}_\pi$ is obtained from data independent of $\mathcal{D}_n$.

C7 Complexity growth condition. The entropy integral of the extended class satisfies $\mathcal{J}(\infty, \mathcal{F}_{\text{ext}}) < \infty$ and $\mathcal{J}(\delta, \mathcal{F}_{\text{ext}}) / \{\delta\sqrt{\log\log(1/\delta)}\} \to \infty$ as $\delta \to 0$.

Condition C4 can be relaxed using weight clipping and appropriate tail conditions on the density ratio and outcomes, following variance-control ideas from emphatic- and importance-weighted reinforcement learning (Hallak et al., 2016; Han & Sung, 2019; Mehrabi & Wager, 2024) and heavy-tailed regression. Condition C6 can similarly be relaxed by standard cross-fitting arguments (Nie & Wager, 2021; Foster & Syrgkanis, 2023). Condition C5 holds automatically when d_π is the stationary density ratio. More generally, it is an overlap condition ensuring that the weighted behavior measure $d_\pi \nu_b$ is sufficiently comparable to μ_π to relate the corresponding L^2-norms. It therefore plays a role analogous to classical concentrability assumptions in our analysis (Xie et al., 2022). Condition C7 controls localized

Rademacher complexity (Bartlett et al., 2005; Munos & Szepesvári, 2008).

To obtain sharper rates under approximate Bellman completeness, we impose the following nonrestrictive Hölder-type relation on $\mathcal{F}$.

C8 Hölder-type sup–L^2 relation. There exist constants $\alpha \in [0,1]$ and $L < \infty$ such that, for all $k \in [K]$, almost surely,

$$\|\mathcal{T}\hat{Q}^{(k)} - \hat{Q}^{(k+1)}\|_\infty \;\le\; L\,\|\mathcal{T}\hat{Q}^{(k)} - \hat{Q}^{(k+1)}\|_{2,\mu_\pi}^{\alpha}.$$

Condition C8 is mild and holds in several standard settings. It holds with $\alpha = 0$ for uniformly bounded classes, and with $\alpha > 0$ when the difference class $\{\mathcal{T}Q - Q' : Q, Q' \in \mathcal{F}\}$ has additional regularity. For finite-dimensional linear models of dimension p, one has $L \asymp \sqrt{p}$ and $\alpha = 1$. For reproducing kernel Hilbert spaces with eigenvalue decay $\lambda_j \asymp j^{-2r}$, one may take $\alpha = \frac{2r}{2r+1}$ (Mendelson & Neeman, 2010, Lemma 5.1). The condition also holds for signed convex hulls of suitable bases (van de Geer, 2014, Lemma 2), and for d-variate Hölder and Sobolev classes of order $s > d/2$ on bounded domains, where $\alpha = 1 - \frac{d}{2s}$ (Bibaut et al., 2021, Lemma 4)(Adams & Fournier, 2003; Triebel, 2006).

Our main technical lemma is stated in terms of the *inherent Bellman error* (Munos & Szepesvári, 2008),

$$\varepsilon_{\text{Bell}} := \sup_{Q \in \mathcal{F}} \|\mathcal{T}(Q) - \mathcal{T}_{\mathcal{F}}(Q)\|_{2,\mu_\pi},$$

which measures the extent to which $\mathcal{F}$ fails to satisfy Bellman completeness in the stationary norm.

Lemma 3 (One-step weighted regression error bound)**.** *Assume C1-C8. There exists a constant $C = C(M, L, c) > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1 - \eta$,*

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k)}) - \hat{Q}^{(k+1)}\|_{2,\mu_\pi} \le\; C\Big(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}}\Big) + C\Big(\Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}^{\frac{1}{1-\alpha}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi} \varepsilon_{\text{Bell}}^{\alpha}\Big).$$

When $\alpha = 0$, we have $1/(1-\alpha) = 1$, so the final two terms reduce to $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_{2,\mu_\pi}$. For $\alpha > 0$, the bound is sharper when the inherent Bellman error $\varepsilon_{\text{Bell}}$ is small, that is, under approximate Bellman completeness. Thus, stronger structure leads to less sensitivity of the one-step error to weight-estimation error. If one instead assumes supremum-norm control of the weight error, then C8 is not needed, and the final two terms can be replaced, up to constants, by $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_\infty\, \varepsilon_{\text{Bell}}$ provided $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_\infty < 1$; see Lemma 10 in the appendix.

For s-sparse linear classes in ambient dimension p, $\delta_n \asymp \sqrt{s\log(p/s)/n}$ (Wainwright, 2019). For VC-subgraph dimension V, $\delta_n \asymp \sqrt{V\log(n/V)/n}$ (Van Der Vaart & Wellner, 1996). For Hölder/Sobolev classes of smoothness s with dimension $d \le 2s$, $\delta_n \asymp n^{-s/(2s+d)}$ (Nickl & Pötscher, 2007; Mendelson & Neeman, 2010).

4.4. Main Finite-Sample Guarantee

Our main result follows by combining the high-probability regression error bound in Lemma 3 with the deterministic inexact Picard iteration bound in Lemma 2.

Theorem 2 (Finite-Sample Error Bound for Stationary-Weighted FQE)**.** *Assume C1–C8. There exists a constant $C = C(M, L, c) > 0$ such that, for all $\eta \in (0,1)$ and $K \in \mathbb{N}$, with probability at least $1 - \eta$,*

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \le \gamma^K\, \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \frac{C}{1-\gamma}\Big(\delta_n + \sqrt{\tfrac{\log(K/\eta)}{n}}\Big) + \frac{C}{1-\gamma}\Big(\Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}^{\frac{1}{1-\alpha}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi} \varepsilon_{\text{Bell}}^{\alpha}\Big).$$

Theorem 2 parallels the classical sup-norm bounds of Munos & Szepesvári (2008), but in the stationary $L^2(\mu_\pi)$ norm and without Bellman completeness. A key difference is the additional density-ratio error term, which in the worst case scales as $\|\hat{d}_\pi/d_\pi - 1\|_{2,\mu_\pi}/(1-\gamma)$ (corresponding to $\alpha = 0$, after absorbing $\varepsilon_{\text{Bell}}$ into constants). The bound is most useful in the consistent regime, where $\|\hat{d}_\pi/d_\pi - 1\|_{2,\mu_\pi} \to 0$. For inconsistent estimators, one can instead analyze weighted FQE under the misspecified weights, possibly with an inflated contraction factor. Theorem 1 further shows that the approximation error of $Q^\star_{\mathcal{F}}$ is bounded, up to a $(1-\gamma)^{-1}$ factor, by the best approximation error of $Q^\star$ in $\mathcal{F}$, namely $\inf_{f \in \mathcal{F}} \|Q^\star - f\|_{2,\mu_\pi}$.

The generic weight-estimation term in Theorem 2 can be made more explicit using guarantees for particular minimax or saddle-point estimators. For example, analyses of minimax estimation in conditional moment models similar to (2) are given by Dikkala et al. (2020); Olivas-Martinez & Rotnitzky (2025), while related minimax estimators for off-policy weighting are studied by Uehara et al. (2020); Wang et al. (2023). At a high level, these results suggest that the weight-estimation term decomposes into

$$\underbrace{\inf_{d \in \mathcal{D}} \|d - d^\dagger\|}_{\text{weight-class approximation}} + \underbrace{\inf_{g \in \mathcal{G}} \|g - g^\dagger\|}_{\text{critic-class approximation}} + \underbrace{\mathrm{stat}(\mathcal{D}, \mathcal{G}, n)}_{\text{statistical error}},$$

where $\mathcal{D}$ and $\mathcal{G}$ are the weight and critic classes, $d^\dagger$ is the target weighting function, and the statistical term is controlled by the localized complexity of these classes; see also

Section 5 of Bennett et al. (2023). Thus, the corresponding term in our bound inherits the usual combination of weight approximation, critic approximation, and statistical error.

Robustness to weight error under approximate Bellman completeness. Theorem 2 shows that small inherent Bellman error $\varepsilon_{\text{Bell}}$ attenuates the effect of weight misspecification in stationary-weighted FQE. In particular, the weight-error contribution is

$$\left\|\frac{\hat{d}_\pi}{d_\pi}-1\right\|_{2,\mu_\pi}^{\frac{1}{1-\alpha}} + \left\|\frac{\hat{d}_\pi}{d_\pi}-1\right\|_{2,\mu_\pi} \varepsilon_{\text{Bell}}^{\alpha}.$$

When $\left\|\frac{\hat{d}_\pi}{d_\pi}-1\right\|_{2,\mu_\pi} < 1$, the first term is smaller than linear in the weight error whenever $\alpha > 0$, while the second is damped by $\varepsilon_{\text{Bell}}^{\alpha}$. As $\alpha \to 1$, the contribution is approximately $\|\frac{\hat{d}_\pi}{d_\pi}-1\|_{2,\mu_\pi}\,\varepsilon_{\text{Bell}}$, so the effect of weight error is small when either the weights are accurate or the inherent Bellman error is small.

Contraction under approximate weights. Our analysis extends beyond exact recovery of the stationary density ratio or other admissible weights. It is enough that the learned weights induce a norm under which $\mathcal{T}$ is contractive, possibly with a larger contraction factor $\overline{\gamma} \in (0,1)$ in place of γ. Thus, even if $\hat{d}_\pi$ converges to a misspecified limit $\overline{d}_\pi$, the resulting weighted norm $L^2(\nu_b \overline{d}_\pi)$ may still improve on the behavior norm used by standard FQE whenever $\mathcal{T}$ contracts in that norm, echoing the motivation behind emphatic weighting in TD learning (Patterson et al., 2022).

A simple sufficient condition is that $\overline{d}_\pi$ be multiplicatively close to the stationary density ratio d_π:

$$(1-\varepsilon)\, d_\pi(s,a) \;\le\; \overline{d}_\pi(s,a) \;\le\; (1+\varepsilon)\, d_\pi(s,a).$$

Then $L^2(\nu_b \overline{d}_\pi)$ and $L^2(\nu_b d_\pi)$ are equivalent, and $\mathcal{T}$ remains a $\overline{\gamma}$-contraction in $L^2(\nu_b \overline{d}_\pi)$, with $\overline{\gamma} \le \gamma\sqrt{\frac{1+\varepsilon}{1-\varepsilon}}$. Thus, small multiplicative errors preserve contraction up to a modest inflation of the contraction factor.

On-policy stability and ergodic limits. Setting $\hat{d}_\pi \equiv 1$ recovers the unweighted FQE update. When the data are drawn on-policy from μ_π, the regression norm is exactly $L^2(\mu_\pi)$, so the update contracts toward $Q^\star_{\mathcal{F}}$. More generally, off-policy degradation is controlled by $\|d_\pi^{-1}-1\|_{2,\mu_\pi}$. Under ergodicity and on-policy trajectory sampling, empirical visitation frequencies converge to μ_π, so the norm mismatch vanishes asymptotically and unweighted FQE inherits the same contraction geometry.

4.5. Implicit weighting by sampling from the stationary distribution

This subsection highlights a complementary regime in which stationary weighting is achieved implicitly through sampling. Suppose P, and hence μ_π, is known. If we can sample $(\tilde{S}_i, \tilde{A}_i, \tilde{S}_i') \sim \mu_\pi \otimes P$ directly, then the design is already matched to the target stationary distribution. The remaining ingredient is the reward, which is not observed under this sampling scheme and must therefore be estimated from the behavior data $\mathcal{D}_n$; let $\hat{r}(s,a)$ denote such an estimate. Standard unweighted FQE can then be run on $\{(\tilde{S}_i, \tilde{A}_i, \hat{r}(\tilde{S}_i, \tilde{A}_i), \tilde{S}_i')\}_{i=1}^n$ with $d_\pi \equiv 1$. We then analyze how reward misspecification affects the resulting estimator $\hat{Q}^{(K)}$.

Define the reward-estimated Bellman operator

$$\hat{\mathcal{T}}_{\mathcal{F}} := \Pi_{\mathcal{F}} \hat{\mathcal{T}}, \qquad \hat{\mathcal{T}} Q := \hat{r} + \gamma\, \pi P Q.$$

Let $\hat{Q}^\star_{\mathcal{F}}$ denote the fixed point of $\hat{\mathcal{T}}_{\mathcal{F}}$, and let $\hat{Q}^{(K)}$ be the K-step FQE iterate based on stationary samples and estimated rewards. Then

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \le \|\hat{Q}^{(K)} - \hat{Q}^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi}.$$

The first term is the usual finite-iteration and statistical error for FQE under stationary sampling, and is controlled by Theorem 2 with $d_\pi \equiv 1$. The second term captures the effect of reward misspecification and is controlled by the following theorem.

Theorem 3 (Reward Misspecification)**.** *Under C1–C2,*

$$\|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma}\,\|\hat{r} - r\|_{2,\mu_\pi}.$$

Moreover, if $\mathcal{F}$ is a linear subspace of $L^2(\mu_\pi)$, then

$$\|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma}\,\|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi}.$$

5. Illustrative Experiments

We illustrate the theory with two controlled experiments that isolate the effect of norm mismatch beyond Bellman completeness. The first uses Baird's classical counterexample to study slowed contraction under off-policy sampling (Baird et al., 1995). The second uses 100-state random sparse Dirichlet (Garnet-style) MDPs to illustrate exponential divergence under norm mismatch (Bhatnagar et al., 2009).

Experiment 1: Stationary weighting restores geometric decay. We consider the 7-state "hub-and-spokes" MDP from Baird et al. (1995) (Appendix A), a canonical counterexample for instability in approximate dynamic programming. The target policy always selects the *solid* action, which deterministically moves the agent to the hub, while the behavior policy selects it with probability $\kappa \in (0,1]$ (otherwise taking the *dashed* action, which moves uniformly to a spoke), so smaller κ corresponds to weaker stationary overlap. We use the standard 7-dimensional linear feature map and run linear FQE, comparing unweighted regression

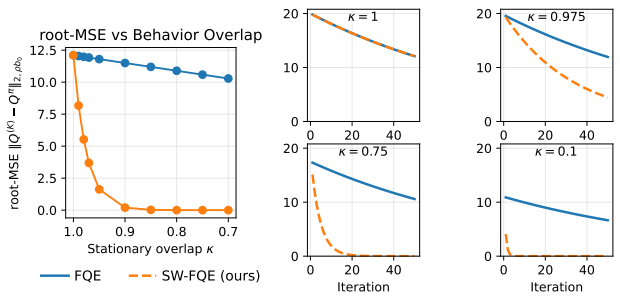

Figure 2. Moderate-overlap regime ($\gamma = 0.95$): root mean squared error (RMSE) at iteration 100 versus κ (left), and iteration curves showing RMSE versus iteration number at various κ (right).

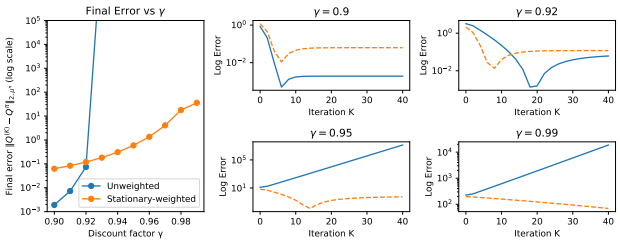

Figure 3. Severe norm-mismatch regime: final error vs. γ (left) and iteration curves (right).

to stationary-weighted regression using the *exact* stationary density ratio. We vary κ from 1 to 0.7 with $\gamma = 0.95$. Errors are reported in the behavior norm $\|Q^{(K)} - Q^\star\|_{2,\nu_b}$ under the empirical behavior distribution.

Figure 2 shows that as overlap decreases, the behavior-norm projection increasingly distorts the Bellman update and weakens its contraction: standard FQE slows dramatically, while SW-FQE maintains nearly uniform geometric decay across all κ. When $\kappa = 1$, the behavior and stationary distributions coincide and the two methods are identical. Results for $\gamma \in \{0.99, 0.999\}$ are deferred to Appendix A.

Experiment 2: Exponential divergence under norm mismatch. We evaluate SW-FQE on random Garnet MDPs with sparse Dirichlet transitions, using $|\mathcal{S}| = 100$ states, $|\mathcal{A}| = 4$ actions, and a fixed branching factor of 5, meaning that for each state–action pair only 5 successor states have nonzero transition probability. The target policy is sampled uniformly at random, while the behavior policy is ε-greedy with $\varepsilon = 0.1$. To amplify norm mismatch, we generate off-policy data using a `reset` sampling scheme that breaks stationarity. We use a low-dimensional linear function class that contains $Q^\star$ exactly but is not Bellman-complete, and compare unweighted FQE to stationary-weighted FQE using the exact stationary density ratios. We sweep $\gamma \in \{0.90, 0.925, 0.95, 0.96, 0.97, 0.98, 0.99\}$, run $K = 200$ fitted iterations, and average over $M = 200$ random seeds. Full details are in Appendix A.

Figure 3 shows the final stationary-norm MSE after K iterations versus γ (left), along with representative error trajectories over iterations (right). It reveals a sharp phase transition in standard FQE as γ increases. For moderate discounts, both methods converge, with error decreasing initially before reaching a shallow *variance elbow*. SW-FQE can incur slightly higher error because stationary weighting amplifies variance by upweighting rare state–action pairs under limited action overlap, whereas unweighted FQE pools and extrapolates. Beyond $\gamma \approx 0.92$, however, unweighted FQE becomes unstable and exhibits clear exponential divergence, even though $Q^\star$ lies exactly in the function class. By contrast, SW-FQE remains stable throughout and continues to exhibit geometric decay in the stationary norm.

Together, these experiments show that standard FQE may remain convergent yet arbitrarily slow under inadequate stationary overlap (Baird), while under severe norm mismatch it can undergo a sharp phase transition and diverge exponentially (Garnet). In both regimes, stationary weighting restores contraction-like behavior and fast geometric decay.

6. Conclusion

This work shows that weighting can stabilize fitted Bellman methods by changing the geometry of each regression update. Standard FVI/FQE fits Bellman regressions under the behavior distribution, even though contraction is governed by the target policy's stationary distribution. Reweighting toward that stationary distribution aligns the regression loss with the contraction norm, yielding a contraction-based analysis and linear convergence beyond Bellman completeness while preserving the simplicity of regression-based FVI/FQE under general function approximation.

This viewpoint links fitted value methods to the emphatic-weighting literature in TD learning, where weighting is used to address off-policy instability. Our contribution is to show that the same idea extends to regression-based FVI/FQE: weighting can be incorporated directly into each Bellman regression step to stabilize fitted updates under misspecification and off-policy sampling. More broadly, this suggests that weighting is a useful design principle for fitted value methods, not only for TD learning.

Important practical challenges remain. Estimating stationary ratios can be difficult, and flexible saddle-point estimators such as DICE may be unstable, especially under weak stationary coverage. A natural next step is to develop weighting schemes that are simpler, more stable, and still improve contraction in fitted Bellman updates. Linear and RKHS-based weight classes may offer a practical alternative to highly flexible ratio estimators while still improving on the unweighted baseline.

Impact Statement.

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Experiments

A.1. Design and additional experiments (Experiment 1)

Experimental setup. We use a modified Baird-style 7-state MDP (Baird et al., 1995). There is a hub state 0 and six spokes $1, \ldots, 6$. The *solid* action deterministically transitions to state 0 from any state, while the *dashed* action transitions uniformly among the spokes. The target policy always chooses the solid action, giving a stationary distribution μ_π that places all its mass on the hub. The behavior policy chooses the solid action with probability $\kappa \in (0, 1]$ and the dashed action otherwise; smaller κ yields weaker stationary overlap because the hub is rarely visited. We use the standard 7-dimensional feature representation from Baird et al. (1995): each spoke has its own basis coordinate, and the hub's feature vector is the sum of all spoke features (plus a bias term). We generate data by simulating transitions under the behavior policy to obtain samples (S_i, A_i, R_i, S_i'), with rewards set to zero so that $Q^\star \equiv 0$. At each FQE iteration we solve a regularized linear least-squares problem in this feature space. For stationary-weighted FQE we use the exact stationary density ratio $d_\pi(a, s) \propto \mu_\pi(a, s)/\nu_b(a, s)$, which is known analytically for this MDP and is nonzero only at the hub, so the experiment isolates the effect of the norm mismatch rather than density-ratio estimation error. All reported metrics are averaged over 100 independent replicates, and errors are computed in the behavior norm $\|\cdot\|_{2,\nu_b}$ using the empirical behavior distribution from each simulated dataset.

A.1.1. Performance under the stationary norm

Figure 4 reports stationary-norm errors for the setting of Experiment 1. As in the main text, decreasing overlap weakens the effective contraction of standard FQE, leading to slow convergence, while SW-FQE continues to exhibit fast geometric decay across all values of κ. Because the stationary norm places all mass on the hub, these curves highlight the dominant error component that unweighted FQE is unable to correct under low overlap.

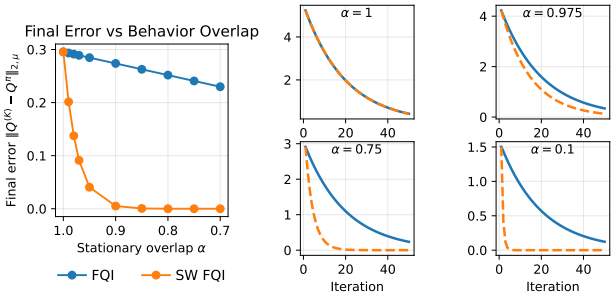

Figure 4. Final stationary-norm error versus stationary overlap for $\gamma = 0.95$. Left: $\|Q^{(K)} - Q^\star\|_{2,\mu_\pi}$ across κ. Right: iteration curves at selected overlap levels.

A.1.2. Larger discount factors

To assess sensitivity to the discount factor, we repeat the stationary-overlap experiment for $\gamma = 0.99$ and $\gamma = 0.999$. As γ increases, the Bellman operator becomes less contractive in the stationary norm, amplifying the effect of norm mismatch. Standard FQE slows substantially in these high-discount regimes, whereas SW-FQE continues to converge rapidly even when overlap is extremely small. These results confirm that the qualitative behavior observed in the main text persists—and becomes more pronounced—as γ approaches 1.

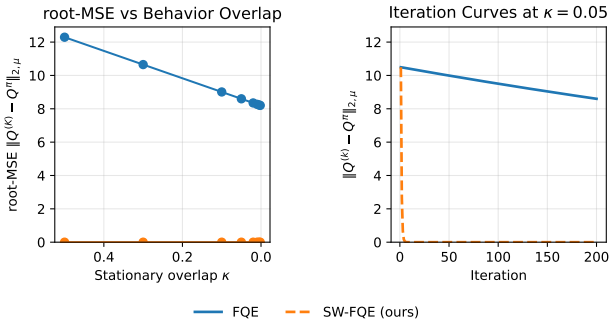

Figure 5. High-discount, low-overlap regime ($\gamma = 0.999$). Left: final error vs. κ. Right: iteration curves at $\kappa = 0.05$.

A.2. Additional details on experiment 2

A.2.1. Experimental design

We evaluate SW-FQE on random Garnet MDPs with $|\mathcal{S}| = 100$, $|\mathcal{A}| = 4$, and branching factor 5. For each state–action pair, 5 distinct next states are sampled uniformly without replacement and assigned transition probabilities via a Dirichlet distribution. Rewards are drawn i.i.d. from $\mathcal{N}(0, 1)$.

For each MDP instance, the target policy π is sampled independently for each state from a Dirichlet distribution over actions. The ground-truth value functions $(V^\pi, Q^\star)$ are computed exactly by solving the Bellman equations via linear system inversion. The stationary distribution d_π is computed by power iteration on the induced Markov chain (Bertsekas & Tsitsiklis, 1996).

The behavior policy μ_π is chosen to be ε-greedy with respect to $Q^\star$ with $\varepsilon = 0.1$, ensuring persistent but limited mismatch with the target. Off-policy datasets of size $n = 10{,}000$ are generated using the `reset` sampling scheme, in which each transition starts from an independent state drawn uniformly from $\mathcal{S}$. This deliberately breaks stationarity and amplifies the norm mismatch between the sampling distribution and the target stationary distribution.

We use a linear Q-function class of dimension $d = 5$ constructed so that $Q^\star$ lies exactly in the span of the features. Specifically, $Q^\star$ is embedded as one column of a random low-dimensional basis over $\mathcal{S} \times \mathcal{A}$, ensuring that the model is *correct* but generically *not Bellman-complete*. Thus, $\mathcal{TF} \not\subseteq \mathcal{F}$ with probability one.

Unweighted FQE is implemented via standard least-squares regression. Stationary-weighted FQE reweights each regression step using the exact stationary density ratio $w(s, a) = \frac{d_\pi(s)\pi(a|s)}{\hat{\rho}_S(s)\mu_\pi(a|s)}$, where $\hat{\rho}_S$ is the empirical state marginal induced by the dataset. Both methods use ridge regularization with $\lambda = 10^{-6}$ and are initialized at $\theta_0 = 0$.

We sweep the discount factor $\gamma \in \{0.90, 0.925, 0.95, 0.96, 0.97, 0.98, 0.99\}$ and run $K = 200$ fitted iterations. All learning curves and final errors are averaged over $M = 50$ independent random seeds for the MDP, policies, and dataset generation. Errors are reported in the stationary norm $\|Q^{(K)} - Q^\star\|_{2,\mu_{\pi^\pi}}$.

A.2.2. Additional experiments

In this experiment, stationary weighting uses *estimated* stationary density ratios obtained by solving the corresponding DualDICE min–max saddle-point problem in closed form on the tabular MDP. This introduces additional variance from ratio estimation, which can slightly weaken early-iteration performance. Nevertheless, as seen in Fig. 6, the qualitative stabilization effect persists: even with estimated weights,

stationary reweighting prevents exponential divergence and preserves geometric decay under severe norm mismatch.

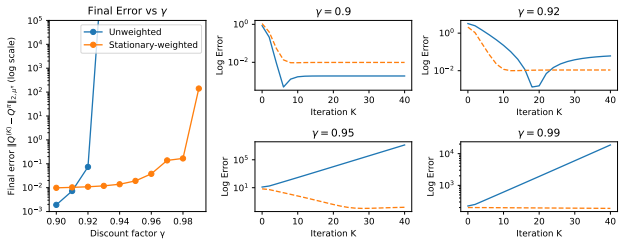

Figure 6. Severe norm-mismatch regime with *estimated* stationary density ratios: final error versus γ (left) and iteration curves (right).

B. Fixed-Point Estimation of Stationary Ratios

The stationary ratio $d_\pi = d\mu_\pi / d\nu_b$ is identified by the fixed-point equation implied by stationarity $\mu_\pi = \mu_\pi P_\pi$:

$$d_\pi(s', a') = (\mathcal{K} d_\pi)(s', a'),$$

where the linear operator $\mathcal{K}$ is

$$(\mathcal{K}g)(s', a') = \frac{\pi(a' \mid s')}{\nu_b(a' \mid s')} \mathbb{E}_{(S,A)\sim\nu_b}[g(S, A)\, k(S, A, s')],$$

and

$$k(s, a, s') = \frac{P(s' \mid s, a)}{\rho_{b,1}(s')}, \rho_{b,1}(s') := \mathbb{E}_{(S,A)\sim\nu_b}[P(s' \mid S, A)].$$

is the transition density ratio, with $\rho_{b,1}$ the marginal density of S' induced by the behavior dynamics. The quantity k can be estimated directly using the empirical squared-loss objective of Hines & Miles (2025):

$$\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} k(S_i, A_i, S'_j)^2 \;-\; \frac{2}{n}\sum_{i=1}^{n} k(S_i, A_i, S'_i).$$

However, $\mathcal{K}$ is the adjoint of πP in $L^2(\mu_\pi)$ and satisfies $\|\mathcal{K}\|_{L^2(\mu_\pi)} = 1$, so direct Picard iteration of $d_\pi = \mathcal{K} d_\pi$ is not contractive. A standard workaround is the discounted resolvent equation

$$d_{\gamma'} = (1-\gamma')\tfrac{\pi}{\nu_b} + \gamma'\, \mathcal{K} w_{\gamma'},$$

whose solution $d_{\gamma'}$ is the γ'-discounted occupancy ratio. The resolvent replaces exact stationarity by a γ'-discounted flow balance, which yields a γ'-contractive update. Since $\mathcal{K}$ is nonexpansive in $L^2(\mu_\pi)$, Picard iteration applies and $w_{\gamma'}$ can be computed analogously to fitted value iteration.

Under standard ergodicity assumptions—irreducibility, aperiodicity, and geometric mixing—the discounted ratios satisfy $w_{\gamma'} \to d_\pi$ as $\gamma' \uparrow 1$ (Meyn & Tweedie, 1993; Glynn & Meyn, 1996). The resolvent formulation induces a familiar bias–variance trade-off: smaller $1-\gamma'$ improves numerical stability but introduces bias of order $O(1-\gamma')$, whereas letting $\gamma' \uparrow 1$ removes the bias at the expense of variance scaling like $(1-\gamma')^{-1}$ due to the effective horizon of the discounted occupancy measure.

C. Contraction Results

C.1. Contraction Under Stationary Measures

Lemma 4 (Bellman contraction for Q-functions under a stationary measure)**.** *If μ_π is stationary for πP, then πP is nonexpansive in $L^2(\mu_\pi)$:*

$$\|\pi P(Q_1 - Q_2)\|_{2,\mu_\pi} \;\le\; \|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Therefore the Bellman operator $\mathcal{T}$ is a γ-contraction:

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \;\le\; \gamma\, \|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Proof. Let $h := Q_1 - Q_2$. For any (s, a), Jensen's inequality gives

$$\big(\pi P h\big)(s, a)^2 \;\le\; \big(\pi P(h^2)\big)(s, a),$$

since πP is a conditional expectation operator.

Integrating with respect to μ_π and using stationarity $\mu_\pi = \mu_\pi(\pi P)$,

$$\|\pi P h\|_{2,\mu_\pi}^2 = \int (\pi P h)^2\, d\mu_\pi \le \int \pi P(h^2)\, d\mu_\pi = \int h^2\, d\mu_\pi.$$

Thus πP is nonexpansive: $\|\pi P h\|_{2,\mu_\pi} \le \|h\|_{2,\mu_\pi}$.

For the Bellman operator,

$$\mathcal{T}Q_1 - \mathcal{T}Q_2 = \gamma\, \pi P(Q_1 - Q_2) = \gamma\, \pi P h.$$

Taking norms and applying nonexpansiveness proves

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \le \gamma \|h\|_{2,\mu_\pi},$$

as claimed. □

Lemma 5 (Contraction of the projected Bellman operator)**.** *If μ_π is stationary for πP, then $\mathcal{T}_\mathcal{F}$ is a γ-contraction on $L^2(\mu_\pi)$; that is, for all $Q_1, Q_2 \in L^2(\mu_\pi)$*

$$\|\mathcal{T}_\mathcal{F} Q_1 - \mathcal{T}_\mathcal{F} Q_2\|_{2,\mu_\pi} \;\le\; \gamma\, \|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Proof. By Lemma 4,

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \le \gamma\, \|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Since $\Pi_\mathcal{F}$ is the $L^2(\mu_\pi)$ orthogonal projection onto a closed convex set, it is nonexpansive:

$$\|\Pi_\mathcal{F} f - \Pi_\mathcal{F} g\|_{2,\mu_\pi} \le \|f - g\|_{2,\mu_\pi}.$$

Thus

$$\begin{aligned}\|\mathcal{T}_{\mathcal{F}}Q_1 - \mathcal{T}_{\mathcal{F}}Q_2\|_{2,\mu_\pi} &= \|\Pi_{\mathcal{F}}(\mathcal{T}Q_1) - \Pi_{\mathcal{F}}(\mathcal{T}Q_2)\|_{2,\mu_\pi} \\ &\le \|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \\ &\le \gamma\,\|Q_1 - Q_2\|_{2,\mu_\pi}.\end{aligned}$$

□

Proof of Lemma 1. The first claim follows directly from Lemma 4. The second claim follows from Lemma 5. □

D. Technical lemmas

D.1. Local maximal inequality

Let $O_1, \dots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \to \mathbb{R}$, define

$$\|f\| := \sqrt{\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[f(O_i)^2]}. \tag{3}$$

For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0, \infty)$, define the localized Rademacher complexity

$$\mathcal{R}_n(\mathcal{F}, \delta) := \mathbb{E}\left[\sup_{\substack{f\in\mathcal{F}\\ \|f\|\le\delta}} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i f(O_i)\right],$$

where ϵ_i are i.i.d. Rademacher random variables.

The following lemma provides a local maximal inequality and restates Lemma 11 of (Foster & Syrgkanis, 2023) (see also Lemma 11 of van der Laan et al. (2025b)).

Lemma 6 (Local maximal inequality)**.** *Let $\mathcal{F}$ be a star-shaped class of functions satisfying $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$. Let $\delta = \delta_n \in (0, 1)$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F}, \delta) \le \delta^2$, and suppose that, as $n \to \infty$,*

$$\frac{1}{\sqrt{n}}\sqrt{\log\log(1/\delta_n)} = o(\delta_n).$$

Then there exists a constant $C > 0$ such that, for all $\eta \in (0, 1)$, with probability at least $1 - \eta$, every $f \in \mathcal{F}$ satisfies

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) &\le C\Big(\delta^2 + \delta\,\|f\|\Big) \\ &+ C\Big(\frac{\sqrt{\log(1/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big).\end{aligned}$$

Proof. Lemma 11 in (van der Laan et al., 2025b) shows that there exists a constant $C > 0$ (possibly depending on the choice of sequence $\{\delta_n\}$) such that, for all $u \ge 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \le C\Big(\delta_n^2 + \delta_n\,\|f\| + \frac{u\,\|f\|}{\sqrt{n}} + \frac{M\,u^2}{n}\Big).$$

Set $u := \sqrt{\log(1/\eta)}$, so that $e^{-u^2} = e^{-\log(1/\eta)} = \eta$. Substituting this choice of u into the above inequality yields, with probability at least $1 - \eta$, every $f \in \mathcal{F}$ satisfies

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) &\le C\Big(\delta_n^2 + \delta_n\,\|f\|\Big) \\ &+ C\Big(\frac{\sqrt{\log(1/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big).\end{aligned}$$

□

The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy integral and is a direct consequence of Theorem 2.1 of Van Der Vaart & Wellner (2011).

For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}$ under the $L^2(Q)$ norm (Van Der Vaart & Wellner, 1996). Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_Q \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))}\, d\epsilon,$$

where the supremum is taken over all discrete probability distributions Q.

Lemma 7. *Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$. Then, for every $\delta > 0$,*

$$\mathcal{R}_n(\mathcal{F}, \delta) \lesssim \frac{1}{\sqrt{n}}\,\mathcal{J}(\delta, \mathcal{F})\left(1 + \frac{\mathcal{J}(\delta, \mathcal{F})}{\delta\sqrt{n}}\right),$$

where the implicit constant depends only on M.

Proof. This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart & Wellner (2011); see in particular the step where the local Rademacher complexity is controlled by the uniform entropy integral for star-shaped classes. □

D.2. Regret of a Single Inexact Picard Step

Recall the weight function d_π in Condition C3, and let $\hat{d}_\pi$ denote an estimate of this weight that is trained independently of $\mathcal{D}_n$ (C6).

Let $\mathcal{F}$ be a convex function class and let $\hat{Q}^{\text{init}} \in \mathcal{F}$ be an initial (possibly data-dependent) estimate. Define the stationary-weighted population risk

$$\hat{R}_0(Q) = \mathbb{E}_{\nu_b}\Big[d_\pi(S, A)\,\big\{R + \gamma(\pi\hat{Q}^{\text{init}})(S') - Q(S, A)\big\}^2\Big].$$

Following Algorithm 1, define the empirical minimizer

$$\hat{Q}_n := \arg\min_{Q \in \mathcal{F}} \hat{R}_n(Q),$$

where

$$\hat{R}_n(Q) = \sum_{i=1}^{n} \hat{d}_\pi(S_i, A_i) \left\{R_i + \gamma(\pi\hat{Q}^{\text{init}})(S_i') - Q(S_i, A_i)\right\}^2.$$

Lemma 8 shows that this stationary-weighted population objective is correctly specified for the projected Bellman operator $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ *only* under the orthogonality condition in Condition C3.

Lemma 8 (Fixed-point operator as population risk minimizer)**.** *Under Conditions C3,*

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) = \arg\min_{Q \in \overline{\mathcal{F}}} \hat{R}_0(Q).$$

Proof. Consider the stationary-population objective

$$\tilde{R}_0(Q) := \mathbb{E}_{\mu_\pi}\left[\left\{R + \gamma(\pi\hat{Q}^{\text{init}})(S') - Q(S, A)\right\}^2 \mid \mathcal{D}_n\right].$$

By definition,

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) = \arg\min_{Q \in \overline{\mathcal{F}}} \tilde{R}_0(Q),$$

as this is precisely the $\mathcal{F}$-projection of the Bellman target $R + \gamma(\pi\hat{Q}^{\text{init}})(S')$.

By the restricted stationary overlap condition in C3, the directional derivatives of $\tilde{R}_0$ and $\hat{R}_0$ coincide. In particular, for every direction $h \in \mathcal{F}$, the first order optiomality conditions for $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ imply that, for $Y^{\text{tgt}} := R + \gamma(\pi\hat{Q}^{\text{init}})(S')$,

$$\begin{aligned} 0 &= \mathbb{E}_{\mu_\pi}\left[h(S, A)\left\{\mathcal{T}(\hat{Q}^{\text{init}})(S, A) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S, A)\right\}\right] \\ &= \mathbb{E}_{\mu_\pi}\left[h(S, A)\left\{Y^{\text{tgt}} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S, A)\right\}\right] \\ &= \mathbb{E}_{\nu_b}\left[d_\pi(S, A)h(S, A)\{Y^{\text{tgt}} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S, A)\}\right]. \end{aligned}$$

Thus $\tilde{R}_0$ and $\hat{R}_0$ have identical first-order optimality conditions over $\overline{\mathcal{F}}$, and therefore the same minimizer.

□

The following results control the empirical process fluctuations that arise in the analysis of the inexact Picard iteration. Recall the extended function class

$$\mathcal{F}_{\text{ext}} := \mathcal{F} \cup \mathcal{F}_\pi,$$

where $\mathcal{F}_\pi := \{(s, a) \mapsto (\pi f)(s) : f \in \mathcal{F}\}$. Recall also the critical radius

$$\delta_n := \inf\left\{\delta > 0 : \frac{\mathcal{J}(\delta, \mathcal{F}_{\text{ext}})}{\sqrt{n}\,\delta^2} \le 1\right\}.$$

Lemma 9. *Let w be a state-action function. Assume that w, R, $\hat{Q}^{\text{init}}$, $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$, and $\hat{Q}_n$ are uniformly bounded by M. Assume further that C6 and C7 hold. Then there exists a constant $C = C(M) > 0$ such that, for all $\eta \in (0, 1)$, with probability at least $1 - \eta$,*

$$\begin{aligned} &\left|(P_n - P_0)\left[w\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\left\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\right\}\right]\right| \\ &\le C\Big(\delta_n^2 + \delta_n \left\|w(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\right\|\Big) \\ &\quad + C\Big(\frac{\sqrt{\log(1/\eta)}\left\|w(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\right\|}{\sqrt{n}} + \frac{\log(1/\eta)}{n}\Big), \end{aligned}$$

where $\|\cdot\|$ denotes the $L^2(P_0)$ behavior norm.

Proof. Define the function class

$$\begin{aligned} \mathcal{G} := \Big\{&(s, a, r, s') \mapsto \big(Q_1(s, a) - Q_2(s, a)\big) \\ &\times \big(r + \gamma V_1(s') - Q_2(s, a)\big) : \; Q_1, Q_2 \in \mathcal{F}, \; V_1 \in \mathcal{F}_\pi\Big\}. \end{aligned}$$

Without loss of generality, assume that $\mathcal{F}$ and $\mathcal{F}_\pi$ are uniformly bounded by M (for example, by restricting these classes to functions satisfying $\|f\|_\infty \le M$). Since $\mathcal{F}$ and $\mathcal{F}_\pi$ are uniformly bounded, each element of $\mathcal{G}$ can be written as a Lipschitz-continuous pointwise transformation of elements of $\mathcal{F} \times \mathcal{F} \times \mathcal{F}_\pi$, with Lipschitz constant depending only on the uniform bounds and γ. Standard entropy bounds for Lipschitz images of function classes (Van Der Vaart & Wellner, 1996) then imply that there exists a constant $C < \infty$ such that, for all $\delta > 0$,

$$\mathcal{J}(\delta, \mathcal{G}) \lesssim \mathcal{J}(\delta, \mathcal{F}) + \mathcal{J}(\delta, \mathcal{F}_\pi) \lesssim 2\mathcal{J}(\delta, \mathcal{F}_{\text{ext}}).$$

By Lemma 7, the critical radius $\delta_n(\mathcal{G})$ is upper bounded by the uniform entropy-based radius δ_n. Moreover, by C7, we have

$$\frac{\mathcal{J}(\delta, \mathcal{F}_{\text{ext}})}{\delta\sqrt{\log\log(1/\delta)}} \longrightarrow \infty \quad \text{as } \delta \to 0,$$

and hence $n^{-1/2}\sqrt{\log\log(1/\delta_n)} = o(\delta_n)$. Thus, applying Lemma 6, there exists a universal constant $C > 0$ such that, for all $\eta \in (0, 1)$, with probability at least $1 - \eta$, every $f \in \mathcal{F}$ satisfies

$$\begin{aligned} &\frac{1}{n}\sum_{i=1}^{n}\big(w(S_i, A_i)f(O_i) - \mathbb{E}[w(S_i, A_i)f(O_i)]\big) \\ &\le C\Big(\delta_n^2 + \delta_n \|wf\|\Big) \\ &\quad + C\Big(\frac{\sqrt{\log(1/\eta)}\,\|wf\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big), \end{aligned}$$

where $O_i = (S_i, A_i, R_i, S_i')$ and $\mathbb{E}_w[f(O)] = \mathbb{E}[w(S, A)f(O)]$. Here we use that the weight function w is uniformly bounded and that multiplication by a fixed

bounded function preserves uniform entropy numbers. Thus, the entropy of the class $\{ wf : f \in \mathcal{G} \}$ is controlled by that of $\mathcal{G}$, up to universal constants and the uniform bound.

□

Main bound for inexact Picard iteration. The following lemma bounds the regret incurred by an inexact Picard iteration in terms of the critical radius of $\mathcal{F}$ and the χ^2-divergence of the density ratio d_π. It allows the starting point $\hat{Q}^{\text{init}}$ to be an arbitrary (possibly data-dependent) element of $\mathcal{F}$.

Define the weighted norm:

$$\|f\|_{2,d_\pi}^2 := P_0\big[d_\pi f^2\big].$$

Lemma 10 (Regret for inexact iteration)**.** *Assume that $\hat{d}_\pi$, d_π, R, $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$, and $\hat{Q}_n$ are uniformly bounded by M. Assume further that C5, C6, and C7 hold. Let $\alpha \in [0,1]$ satisfy C8. Then there exists a constant $C = C(M, c, L) > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$,*

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\quad \le C\Big(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi}^{\frac{1}{1-\alpha}} \\ &\quad + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi} \big\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\big\|_{2,d_\pi}^{\alpha}\Big).\end{aligned} \tag{4}$$

for some constant $C > 0$. Furthermore, without C8, it holds that

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\quad \le C\Bigg(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} \\ &\qquad + C\Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{\infty} \big\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\big\|_{2,d_\pi}\Bigg).\end{aligned} \tag{5}$$

for some constant $C > 0$, provided $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_\infty < 1$.

Proof of Lemma 10. By first-order optimality of $\hat{Q}^\star = \arg\min_{Q \in \mathcal{F}} \hat{R}_n(Q)$ and convexity of $\mathcal{F}$, we have, for all $Q \in \mathcal{F}$,

$$\begin{aligned}&\frac{1}{n}\sum_{i=1}^n \hat{d}_\pi(S_i, A_i)\left\{Q(S_i, A_i) - \hat{Q}^\star(S_i, A_i)\right\} \\ &\qquad \times \left\{R_i + \gamma\left(\pi\hat{Q}^{\text{init}}\right)(S_i') - \hat{Q}^\star(S_i, A_i)\right\} \;\le\; 0.\end{aligned}$$

Note that $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) \in \mathcal{F}$ by definition of the projected Bellman operator $\mathcal{T}_{\mathcal{F}}$. Taking $Q = \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n$ in the empirical first-order condition gives

$$\begin{aligned}&\frac{1}{n}\sum_{i=1}^n \hat{d}_\pi(S_i, A_i)\left\{\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S_i, A_i) - \hat{Q}_n(S_i, A_i)\right\} \\ &\qquad \times \left\{R_i + \gamma\left(\pi\hat{Q}^{\text{init}}\right)(S_i') - \hat{Q}_n(S_i, A_i)\right\} \le 0.\end{aligned}$$

Adding and subtracting the population expectation and rearranging,

$$\begin{aligned}&P_0\big[\hat{d}_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \le \\ &(P_0 - P_n)\big[\hat{d}_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big].\end{aligned} \tag{6}$$

Here we abuse notation and write R for the coordinate projection $(s, a, r, s') \mapsto r$, and similarly view $(\pi\hat{Q}^{\text{init}})$ as the mapping $(s, a, r, s') \mapsto (\pi\hat{Q}^{\text{init}})(s')$.

By the law of total expectations and Adding and subtracting d_π, we obtain

$$\begin{aligned}&P_0\big[\hat{d}_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \\ &= P_0\big[\hat{d}_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \\ &= \quad P_0\big[d_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \\ &\qquad + P_0\big[(\hat{d}_\pi - d_\pi)\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big].\end{aligned}$$

Applying the law of total expectation,

$$\begin{aligned}&P_0\Big[\hat{d}_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\Big] \\ &= P_0\Big[d_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\left(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\Big] \\ &\quad + P_0\Big[(\hat{d}_\pi - d_\pi)\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\}\Big].\end{aligned}$$

The first-order optimality conditions for $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ as a population minimizer (Lemma 8) imply, for all $\hat{Q}^{\text{init}} \in \mathcal{F}$,

$$P_0\big[d_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\}\big] \;\ge\; 0.$$

Thus

$$\begin{aligned}&P_0\Big[d_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\left(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\Big] \\ &= P_0\Big[d_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\Big] \\ &\quad + P_0\Big[d_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\left(\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\right)\Big] \\ &\ge \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^2.\end{aligned}$$

By Cauchy–Schwarz,

$$\begin{aligned}&P_0\big[(\hat{d}_\pi - d_\pi)\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \\ &= P_0\big[d_\pi\big(\tfrac{\hat{d}_\pi}{d_\pi} - 1\big)\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \\ &\ge -\Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi} - 1\big)\big(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big)\Big\|_{2,d_\pi} \big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big\|_{2,d_\pi}\end{aligned}$$

where $\|f\|_{2,d_\pi}^2 := P_0\big[d_\pi f^2\big]$.

Combining the above, we obtain the local quadratic lower bound

$$\begin{aligned}&P_0\big[\hat{d}_\pi\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \\ &\ge \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^2 \\ &\quad - \Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi} - 1\big)\big(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big)\Big\|_{2,d_\pi} \big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big\|_{2,d_\pi}.\end{aligned}$$

Next, applying this lower bound to (6) and rearranging,

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^2 \\ &\le (P_0 - P_n)\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\,\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big] \\ &+ \Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi} - 1\big)(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\Big\|_{2,d_\pi} \big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big\|_{2,d_\pi}.\end{aligned} \tag{7}$$

Denoting the Bellman residual by

$$\xi_n := R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n,$$

the empirical process term satisfies

$$\begin{aligned}&(P_0 - P_n)\Big[\hat{d}_\pi\,\big(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big)\,\xi_n\Big] \\ &= (P_0 - P_n)\Big[d_\pi\,\big(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big)\,\xi_n\Big] \\ &+ (P_0 - P_n)\Big[\big(\hat{d}_\pi - d_\pi\big)\,\big(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big)\,\xi_n\Big].\end{aligned}$$

We now invoke C6 and apply Lemma 9 conditionally on the data used to train the weights to the two terms on the right-hand side, with $w = d_\pi$ and $w = \hat{d}_\pi - d_\pi$, both of which are uniformly bounded by Assumption C4. This yields a constant $C > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$,

$$\begin{aligned}&\Big|(P_0 - P_n)\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\,\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big]\Big| \\ &\le C\Big(\delta_n^2 + \delta_n\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}\Big) \\ &+ C\left(\frac{\sqrt{\log(1/\eta)}\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}}{\sqrt{n}} + \frac{\log(1/\eta)}{n}\right).\end{aligned}$$

Combining this high-probability bound with (7), we conclude

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^2 \\ &\le C\Big(\delta_n^2 + \delta_n\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}\Big) \\ &+ C\left(\frac{\sqrt{\log(1/\eta)}\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}}{\sqrt{n}} + \frac{\log(1/\eta)}{n}\right) \\ &+ \Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi} - 1\big)(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\Big\|_{2,d_\pi} \big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big\|_{2,d_\pi}.\end{aligned} \tag{8}$$

The inequality above, together with the loose bound

$$\begin{aligned}&\big\|(\tfrac{\hat{d}_\pi}{d_\pi} - 1)(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\big\|_{2,d_\pi} \\ &\lesssim \big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\big\|_{2,d_\pi}\end{aligned} \tag{9}$$

implies that, with probability at least $1-\eta$,

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\quad\le C\left(\delta_n + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi} + \sqrt{\frac{\log(1/\eta)}{n}}\right),\end{aligned} \tag{10}$$

for some constant C.

To obtain a sharper bound, we invoke Condition C8 and C5: there exists some $\alpha \in [0,1]$ such that

$$\|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_\infty \;\le\; cL\|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^\alpha.$$

Then, by Hölder's inequality, the bound in (9) can be improved to

$$\begin{aligned}&\Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi} - 1\big)\big(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big)\Big\|_{2,d_\pi} \\ &\lesssim \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi} \times \|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^\alpha.\end{aligned}$$

We can then apply Lemma 9 together with (7) and argue as before to conclude that, with probability at least $1-\eta$,

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\le C\Big(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi}^{\frac{1}{1-\alpha}} \\ &\quad + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi} \big\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\big\|_{2,d_\pi}^\alpha\Big).\end{aligned} \tag{11}$$

for some constant $C > 0$. The first claim then follows.

Bound with sup-norm weight error. If one is willing to measure weight accuracy in supremum norm, an alternative bound for (9) is

$$\begin{aligned}&\Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi} - 1\big)\big(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\big)\Big\|_{2,d_\pi} \\ &\quad\le \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_\infty \|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\quad\le \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_\infty \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\qquad + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_\infty \|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\|_{2,d_\pi}.\end{aligned}$$

In that case, we obtain

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\le C\Bigg(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} \\ &\qquad + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_\infty \big\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\big\|_{2,d_\pi}\Bigg).\end{aligned} \tag{12}$$

for some constant $C > 0$, provided $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_\infty < 1$.

□

E. Proofs of Main results

Proof of Theorem 1. Write

$$\begin{aligned}Q_{\mathcal{F}}^\star - Q^\star &= \mathcal{T}_{\mathcal{F}}Q_{\mathcal{F}}^\star - \mathcal{T}Q^\star \\ &= \big(\mathcal{T}_{\mathcal{F}}Q_{\mathcal{F}}^\star - \mathcal{T}_{\mathcal{F}}Q^\star\big) + \big(\mathcal{T}_{\mathcal{F}}Q^\star - \mathcal{T}Q^\star\big).\end{aligned}$$

Taking $L^2(\mu_\pi)$ norms and applying the triangle inequality

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \|\mathcal{T}_{\mathcal{F}} Q^\star_{\mathcal{F}} - \mathcal{T}_{\mathcal{F}} Q^\star\|_{2,\mu_\pi} + \|\mathcal{T}_{\mathcal{F}} Q^\star - \mathcal{T} Q^\star\|_{2,\mu_\pi}.$$

By the contraction property of Lemma 5,

$$\|\mathcal{T}_{\mathcal{F}} Q^\star_{\mathcal{F}} - \mathcal{T}_{\mathcal{F}} Q^\star\|_{2,\mu_\pi} \le \gamma \, \|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi},$$

and by definition of $\mathcal{T}_{\mathcal{F}}$,

$$\mathcal{T}_{\mathcal{F}} Q^\star - \mathcal{T} Q^\star = \Pi_{\mathcal{F}} \mathcal{T} Q^\star - \mathcal{T} Q^\star = -(I - \Pi_{\mathcal{F}}) \, \mathcal{T} Q^\star.$$

Combining these two displays yields

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \gamma \, \|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} + \left\|(I - \Pi_{\mathcal{F}}) \, \mathcal{T} Q^\star\right\|_{2,\mu_\pi}.$$

Rearranging, we obtain

$$(1-\gamma) \, \|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \left\|(I - \Pi_{\mathcal{F}}) \, \mathcal{T} Q^\star\right\|_{2,\mu_\pi},$$

which proves the first inequality:

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma} \left\|(I - \Pi_{\mathcal{F}}) \, \mathcal{T} Q^\star\right\|_{2,\mu_\pi}.$$

For the second inequality, note that $\Pi_{\mathcal{F}} \mathcal{T} Q^\star$ is the best $L^2(\mu_\pi)$ approximation to $\mathcal{T} Q^\star$ in $\mathcal{F}$, so

$$\left\|(I - \Pi_{\mathcal{F}}) \, \mathcal{T} Q^\star\right\|_{2,\mu_\pi} = \inf_{f \in \mathcal{F}} \|\mathcal{T} Q^\star - f\|_{2,\mu_\pi}.$$

Since $Q^\star$ is a fixed point of $\mathcal{T}$, we have $\mathcal{T} Q^\star = Q^\star$, and hence

$$\left\|(I - \Pi_{\mathcal{F}}) \, \mathcal{T} Q^\star\right\|_{2,\mu_\pi} = \inf_{f \in \mathcal{F}} \|Q^\star - f\|_{2,\mu_\pi}.$$

Substituting this into the previous display yields

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma} \inf_{f \in \mathcal{F}} \|f - Q^\star\|_{2,\mu_\pi},$$

which completes the proof. □

Proof of Lemma 2. By Lemma 5, $\mathcal{T}_{\mathcal{F}}$ is a γ-contraction on $L^2(\mu_\pi)$. Let

$$a_k \;:=\; \|\hat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi}, \qquad k \ge 0,$$

with $\hat{Q}^{(0)} = \hat{Q}$. For any $k \ge 1$,

$$\begin{aligned} a_k &= \|\hat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \\ &\le \|\hat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)})\|_{2,\mu_\pi} + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \\ &\le \eta_k + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - \mathcal{T}_{\mathcal{F}}(Q^\star_{\mathcal{F}})\|_{2,\mu_\pi} \\ &\le \eta_k + \gamma \, \|\hat{Q}^{(k-1)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} = \eta_k + \gamma \, a_{k-1}, \end{aligned}$$

where we used the assumed approximation bound in the first term, the fixed-point property $\mathcal{T}_{\mathcal{F}} Q^\star_{\mathcal{F}} = Q^\star_{\mathcal{F}}$ in the second line, and the contraction property in the third.

Thus the sequence $\{a_k\}$ satisfies the recursion

$$a_k \le \gamma \, a_{k-1} + \eta_k, \qquad k \ge 1.$$

Unrolling this recursion yields

$$\begin{aligned} a_K &\le \gamma^K a_0 + \sum_{j=1}^{K} \gamma^{K-j} \eta_j \\ &= \gamma^K \|\hat{Q} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \; + \; \sum_{j=1}^{K} \gamma^{K-j} \eta_j, \end{aligned}$$

which gives the desired bound. □

Proof of Lemma 3. Under the stated conditions, the claim follows directly from the first part of Lemma 10, using Condition C5 to replace the weighted norm with the $L^2(\mu_\pi)$ norm. □

Proof of Theorem 2. By Lemma 3 and a union bound, there exists a constant $C = C(M) > 0$ such that, for all $\eta \in (0,1)$ and all $k \in [K]$, with probability at least $1 - \eta$,

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - \hat{Q}^{(k)}\|_{2,\mu_\pi} \le C \left(\delta_n + \left\| \tfrac{\hat{d}_\pi}{d_\pi} - 1 \right\|_{2,\mu_\pi} \right) + \; C \sqrt{\frac{\log(K/\eta)}{n}}.$$

Define, for each k,

$$\varepsilon_k := C \left(\delta_n + \left\| \tfrac{\hat{d}_\pi}{d_\pi} - 1 \right\|_{2,\mu_\pi} + \sqrt{\frac{\log(K/\eta)}{n}} \right),$$

so that on this event,

$$\|\hat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)})\|_{2,\mu_\pi} \le \varepsilon_k \quad \text{for all } k \in [K].$$

Thus, by Lemma 2, for any $K \ge 1$,

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \le \gamma^K \, \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \sum_{j=1}^{K} \gamma^{K-j} \, \varepsilon_j.$$

Since ε_j does not depend on j, write $\varepsilon_j = \varepsilon$ for all $j \in [K]$. Hence,

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \le \gamma^K \, \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \; + \; \varepsilon \, \frac{1-\gamma^K}{1-\gamma}.$$

Thus,

$$\begin{aligned} \|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} &\le \gamma^K \, \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \\ &\quad + \frac{C}{1-\gamma} \left(\delta_n + \left\| \tfrac{\hat{d}_\pi}{d_\pi} - 1 \right\|_{2,\mu_\pi} \right) \\ &\quad + \frac{C}{1-\gamma} \left(\sqrt{\frac{\log(K/\eta)}{n}} \right). \end{aligned}$$

□

We proof a more general version of Theorem 3.

Theorem 4 (Reward Misspecification)**.** *Under C1–C2,*

$$\|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma}\,\|\hat{r} - r\|_{2,\mu_\pi}.$$

Moreover, if $\mathcal{F}$ is a linear subspace of $L^2(\mu_\pi)$, then

$$\|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma}\,\|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi}.$$

Hence, for linear classes, only the component $\Pi_{\mathcal{F}}(\hat{r} - r)$ of the reward error affects the fixed point.

Proof of Theorem 3. By Lemma 5 and stationarity of μ_π, $\mathcal{T}_{\mathcal{F}}$ is a γ-contraction on $L^2(\mu_\pi)$.

By definition of the fixed points,

$$Q^\star = \mathcal{T}_{\mathcal{F}}(Q^\star), \qquad \hat{Q}^\star = \hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star).$$

Write

$$\hat{Q}^\star - Q^\star = \hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star),$$

and add and subtract $\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)$:

$$\hat{Q}^\star - Q^\star = \underbrace{\big(\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\big)}_{(I)} + \underbrace{\big(\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star)\big)}_{(II)}.$$

Taking $L^2(\mu_\pi)$-norms and using the triangle inequality,

$$\begin{aligned}\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le\; & \|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\|_{2,\mu_\pi} \\ & + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu_\pi}.\end{aligned}$$

The second term is controlled by the contraction property:

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu_\pi} \le \gamma\,\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi}.$$

Hence

$$(1-\gamma)\,\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le \|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\|_{2,\mu_\pi}. \quad (13)$$

Next, observe that for any Q,

$$\begin{aligned}\|\hat{\mathcal{T}}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q)\|_{2,\mu_\pi} &= \|\Pi_{\mathcal{F}}\hat{\mathcal{T}}(Q) - \Pi_{\mathcal{F}}\mathcal{T}(Q)\|_{2,\mu_\pi} \\ &\le \|\hat{\mathcal{T}}(Q) - \mathcal{T}(Q)\|_{2,\mu_\pi} \\ &= \|\hat{r} - r\|_{2,\mu_\pi},\end{aligned}$$

since $\hat{\mathcal{T}}(Q) - \mathcal{T}(Q) = \hat{r} - r$ does not depend on Q. Substituting into (13) yields

$$\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma}\,\|\hat{r} - r\|_{2,\mu_\pi}.$$

Stronger guarantee under linear function classes. If $\mathcal{F}$ is a linear subspace of $L^2(\mu_\pi)$, then $\Pi_{\mathcal{F}}$ is a linear orthogonal projection, and we obtain a sharper reward misspecification bound. In this case, for any Q,

$$\begin{aligned}\|\hat{\mathcal{T}}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q)\|_{2,\mu_\pi} &= \|\Pi_{\mathcal{F}}\hat{\mathcal{T}}(Q) - \Pi_{\mathcal{F}}\mathcal{T}(Q)\|_{2,\mu_\pi} \\ &= \|\Pi_{\mathcal{F}}\big(\hat{\mathcal{T}}(Q) - \mathcal{T}(Q)\big)\|_{2,\mu_\pi} \\ &= \|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi},\end{aligned}$$

since $\hat{\mathcal{T}}(Q) - \mathcal{T}(Q) = \hat{r} - r$ does not depend on Q. In particular,

$$\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) = \Pi_{\mathcal{F}}(\hat{r} - r),$$

so

$$\|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\|_{2,\mu_\pi} = \|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi}.$$

Substituting this into (13) yields

$$\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma}\,\|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi},$$

so only the component of the reward error lying in $\mathcal{F}$ affects the fixed point.

□